

IFNg condition comparison - Fig7 - Supplementary file 3

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Warning: le package 'knitr' a été compilé avec la version R 4.4.3

Load libraries and functions

```
library(grid)
library(ggplot2)
library(reshape2)
library(UniProt.ws)
library(dplyr)
library(tidyr)
library(VennDiagram)
library(plotly)
library(ggrepel)
library(writexl)

# Connect to UniProt database (human, taxon 9606)
up <- UniProt.ws(taxId = 9606)

# Maps UniProt accessions to gene names and filters
# peptides belonging to CTA genes
map_genes_to_peptides <- function(df_summary) {
  # Expand rows so each protein accession is on its own
  # line
  df_sep <- df_summary %>%
    mutate(RowID = row_number()) %>%
    separate_rows(Prot_accessions, sep = ";\s*") %>%
    mutate(Prot_accessions = sub("-\\d+$", "", Prot_accessions))

  # Query UniProt to retrieve primary gene names
  gene_mapping <- AnnotationDbi::select(up, keys = unique(df_sep$Prot_accessions),
    columns = "gene_primary", keytype = "UniProtKB")

  # Merge gene names back and collapse per peptide
  # sequence
  df_merged <- df_sep %>%
    left_join(gene_mapping, by = c(Prot_accessions = "Entry"))

  df_final <- df_merged %>%
    group_by(Seq) %>%
    summarise(Proteins = paste(unique(Prot_accessions), collapse = "; "),
      Genes = paste(unique(Gene.Names..primary.), collapse = "; "),
      .groups = "drop")

  # Join back to the original dataframe and keep only CTA
  # peptides
  df_summary %>%
    left_join(df_final, by = "Seq") %>%
    filter(Genes %in% l_CTA)
}
```

Load data

```
# Load CTA gene list
l_CTA <- read.table("../data/CTA_list_clean.txt")$V1

# Load full immunopeptidomic dataset (IFNg experiment, both
# cell lines)
df_complete <- read.csv("../data/immunopep_all_clean.tsv", sep = "\t",
  header = T)

# Keep only canonical MHC-I peptides (length 8-11 aa)
df_complete <- df_complete[df_complete$Sequence.Length >= 8 &
  df_complete$Sequence.Length <= 11, ]

# Keep only canonical sequences (exclude non-canonical or
# ambiguous entries)
df_complete <- df_complete[df_complete$Marked.as %in% "Cannonic" |
  df_complete$Marked.as %in% "Cannonic;Mut", ]

# Load NetMHCpan affinity prediction results (all alleles)
df_aff <- read.table("../data/table_res_netmhc_all.tsv", sep = "\t",
  header = T)

# Load IEDB MHC-I reference epitopes
df_iedb_mhc1 <- read.csv("../data/epitope_table_export_human_MHC_I.tsv",
  header = T, sep = "\t")
df_pep_mhc1 <- df_iedb_mhc1[, c("Name", "Source.Molecule.IRI",
  "Molecule.Parent.IRI")]

# Extract clean peptide sequences from IEDB (remove
# modifications and adducts)
pep_iedb_mhc1 <- unique(df_pep_mhc1$Name)
pep_iedb_mhc1 <- gsub(".*|\\+.*", "", pep_iedb_mhc1)
```

I. MHC-I differential analysis (IFNg vs NoIFNg)

1) All peptides volcano plot

CH2879

```
# Prepare data
df_ch_all <- data.frame(pep = df_complete$Annotated.Sequence,
  log2_ratio = log2(df_complete$Abundance.Ratio...CH2879..INF....CH2879..NoINF.),
  log10_p = -log10(df_complete$Abundance.Ratio.Adj..P.Value...CH2879..INF....CH2879..NoINF.),
  Contaminant = df_complete$Contaminant)

# Remove rows with missing values
df_ch_all <- na.omit(df_ch_all)

# Categorise peptides
```

```
df_ch_all$category <- ifelse(df_ch_all$log2_ratio > 1 & df_ch_all$log10_p >
  -log10(0.05), "Upregulated", ifelse(df_ch_all$log2_ratio <
  -1 & df_ch_all$log10_p > -log10(0.05), "Downregulated", "Not significant"))
table_sig <- table(df_ch_all$category)

# Volcano plot
p <- ggplot(df_ch_all, aes(x = log2_ratio, y = log10_p)) + geom_point(aes(color = category),
  alpha = 0.7, size = 1) + scale_color_manual(values = c(Upregulated = "red",
  Downregulated = "blue")) + geom_vline(xintercept = c(-1,
  1), linetype = "dashed") + geom_hline(yintercept = -log10(0.05),
  linetype = "dashed", color = "black") + labs(x = "Log2 Ratio",
  y = "-log10(adj. p-value)", color = "") + theme_minimal(base_size = 13) +
  theme(legend.position = "right") + annotate("text", x = 3,
  y = max(df_ch_all$log10_p, na.rm = TRUE) + 2, label = table_sig["Upregulated"],
  hjust = 1, vjust = 1, size = 4) + annotate("text", x = -3,
  y = max(df_ch_all$log10_p, na.rm = TRUE) + 2, label = table_sig["Downregulated"],
  hjust = 0, vjust = 1, size = 4)
```

p

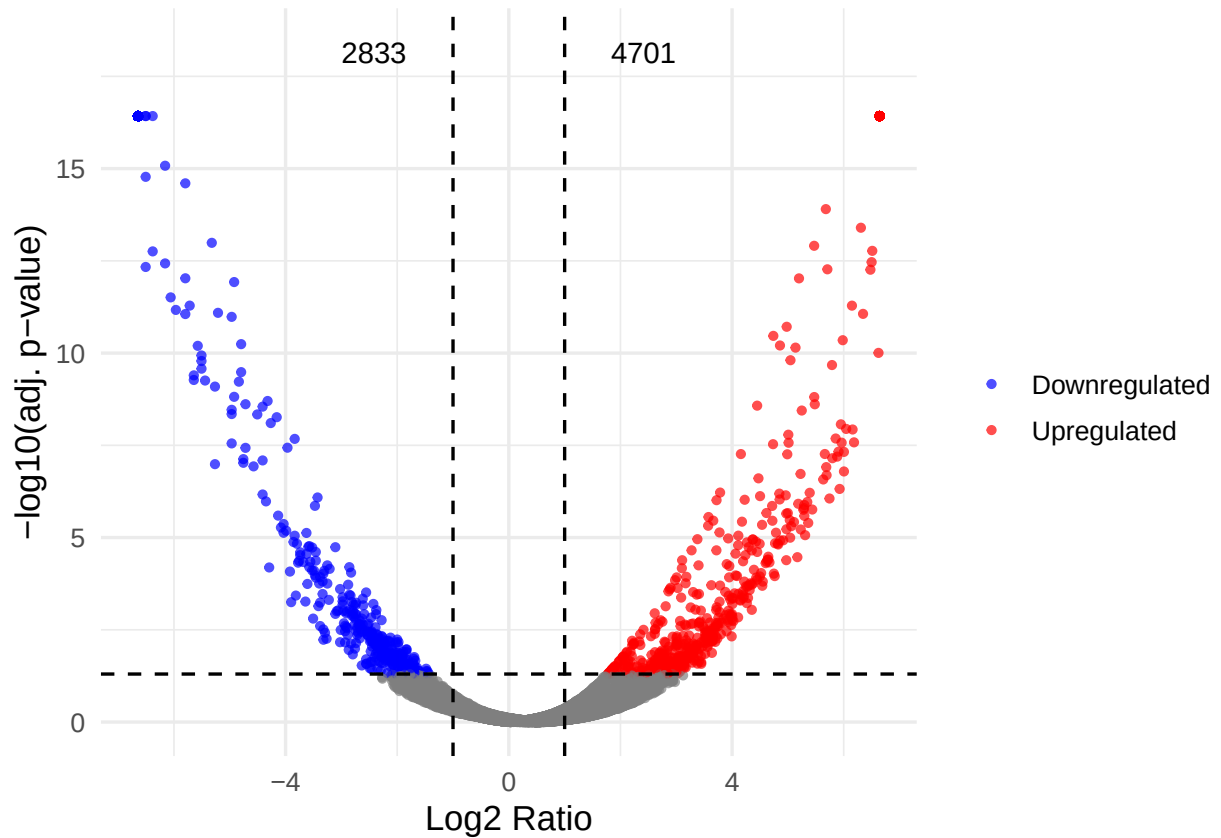


Figure 1: Volcano plot all CH2879

```
ggsave("../results/v_plot_mhc1_ch.svg", plot = p, width = 5,
  height = 5)
```

JJ012

```
# Prepare data
df_jj_all <- data.frame(pep = df_complete$Annotated.Sequence,
  genes = df_complete$Master.Protein.Accessions, log2_ratio = log2(df_complete$Abundance.Ratio...JJ012),
  log10_p = -log10(df_complete$Abundance.Ratio.Adj..P.Value...JJ012..INF.....JJ012..NoINF.))

df_jj_all <- na.omit(df_jj_all)

# Categorise peptides
df_jj_all$category <- ifelse(df_jj_all$log2_ratio > 1 & df_jj_all$log10_p >
  -log10(0.05), "Upregulated", ifelse(df_jj_all$log2_ratio <
  -1 & df_jj_all$log10_p > -log10(0.05), "Downregulated", "Not significant"))
table_sig <- table(df_jj_all$category)

# Volcano plot
p <- ggplot(df_jj_all, aes(x = log2_ratio, y = log10_p)) + geom_point(aes(color = category),
  alpha = 0.7, size = 1) + scale_color_manual(values = c(Upregulated = "red",
  Downregulated = "blue")) + geom_vline(xintercept = c(-1,
  1), linetype = "dashed") + geom_hline(yintercept = -log10(0.05),
  linetype = "dashed", color = "black") + labs(x = "Log2 Ratio",
  y = "-log10(adj. p-value)", color = "") + theme_minimal(base_size = 13) +
  theme(legend.position = "right") + annotate("text", x = 3,
  y = max(df_jj_all$log10_p, na.rm = TRUE) + 2, label = table_sig["Upregulated"],
  hjust = 1, vjust = 1, size = 4) + annotate("text", x = -3,
  y = max(df_jj_all$log10_p, na.rm = TRUE) + 2, label = table_sig["Downregulated"],
  hjust = 0, vjust = 1, size = 4)
p

ggsave("../results/v_plot_mhc1_jj.svg", plot = p, width = 5,
  height = 5)
```

Gene annotation of all peptides

```
# Expand protein accessions (one per row) and strip isoform
# suffixes
df_sep <- df_complete %>%
  mutate(RowID = row_number()) %>%
  separate_rows(Master.Protein.Accessions, sep = ";\s*") %>%
  mutate(Master.Protein.Accessions = sub("-\\d+$", "", Master.Protein.Accessions))

# Query UniProt to get primary gene names for all
# accessions
up <- UniProt.ws(taxId = 9606)

gene_mapping <- AnnotationDbi::select(up, keys = df_sep$Master.Protein.Accessions,
  columns = "gene_primary", keytype = "UniProtKB")

# Merge gene names and collapse multiple genes per peptide
df_merged <- df_sep %>%
```

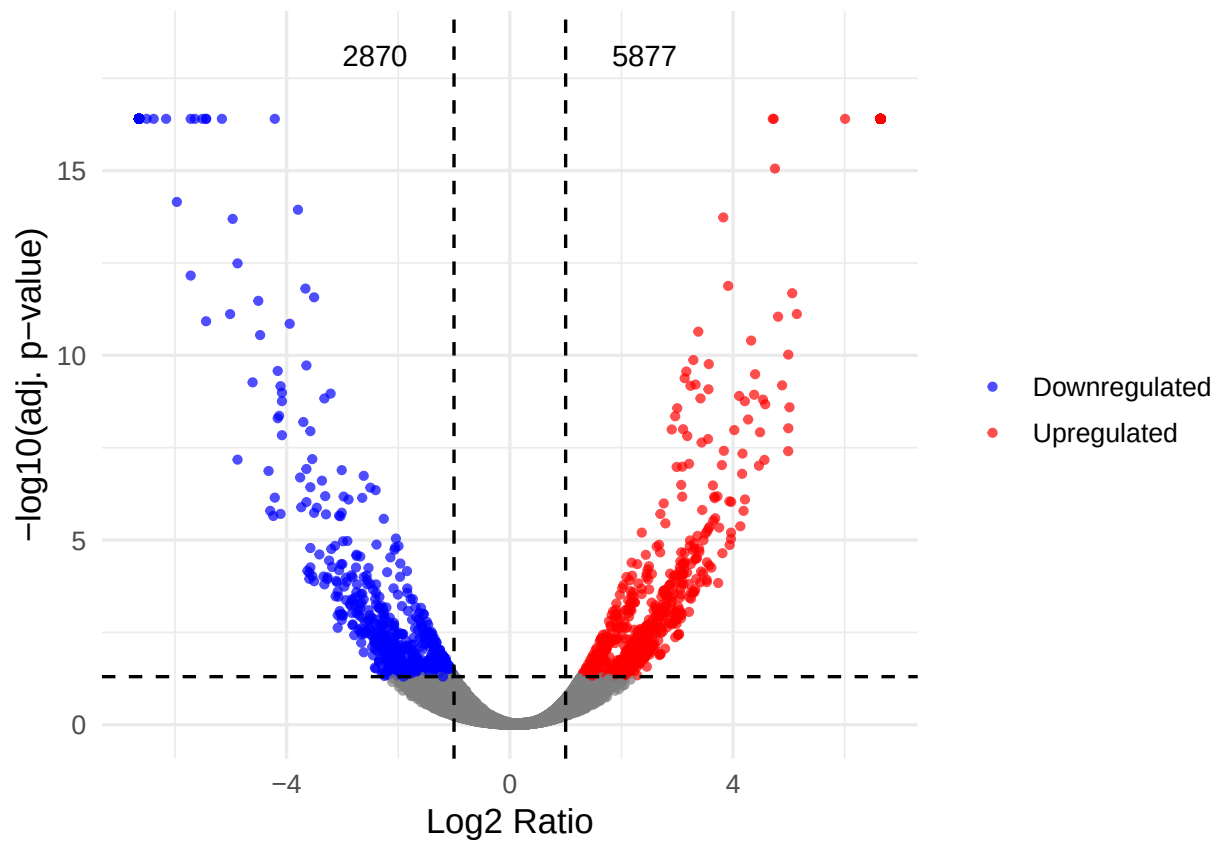


Figure 2: Volcano plot all JJ012

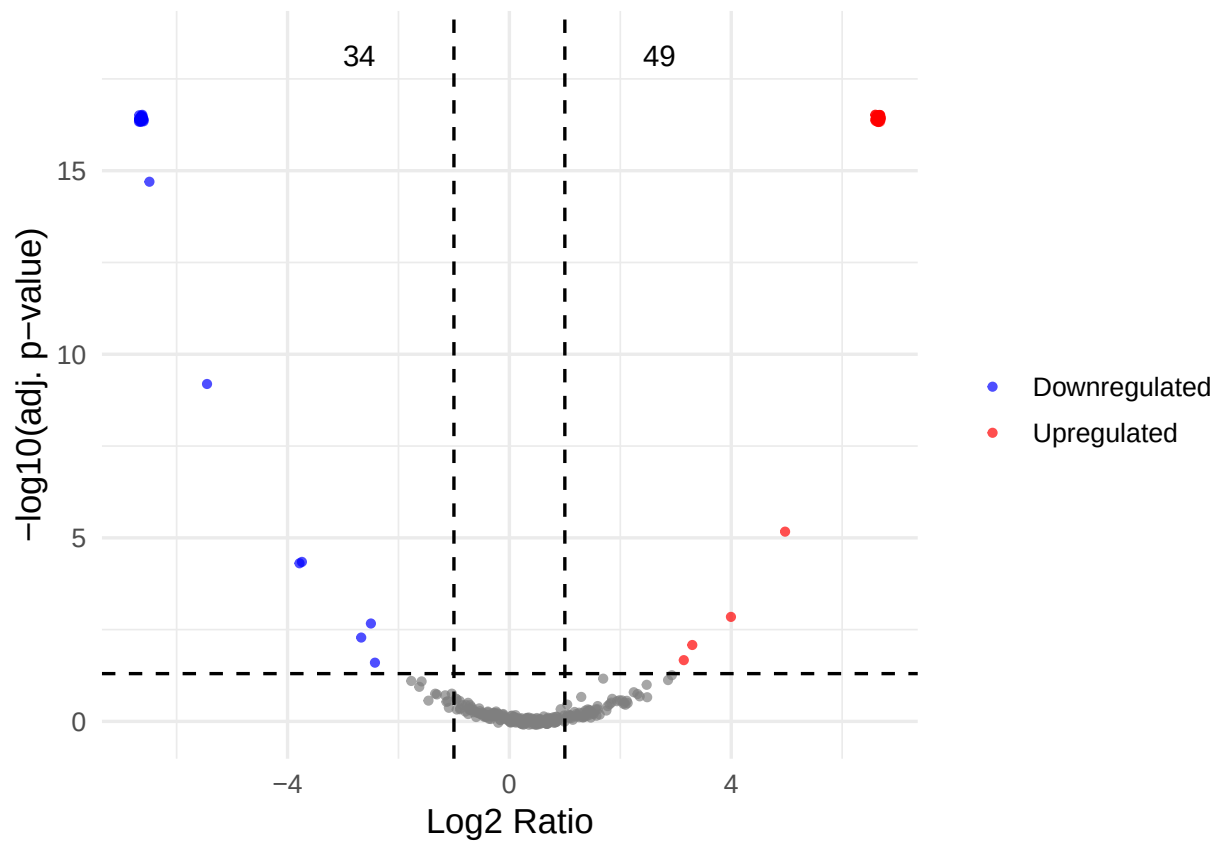


Figure 3: Volcano plot CTAs CH2879

```
# Extract and format upregulated CTA peptides for CH2879
df_up_ch <- df_ch_cta %>%
  filter(category == "Upregulated") %>%
  arrange(desc(log2_ratio))

df_up_ch$Peptide <- gsub("^.*\\.\\.(.*)\\.\\..*$", "\\1", df_up_ch$pep)
df_up_ch$pvalue <- 10^(-df_up_ch$log10_p)
df_up_ch$Gene <- df_up_ch$Genes

df_up_ch <- df_up_ch[, c("Gene", "Peptide", "log2_ratio", "pvalue")]
```

```
# Build volcano dataframe restricted to CTA-derived
# peptides (JJ012)
df_jj_cta <- data.frame(Genes = df_final_pep_cta$Genes, pep = df_final_pep_cta$Annotated.Sequence,
  log2_ratio = log2(df_final_pep_cta$Abundance.Ratio...JJ012..INF....JJ012..NoINF.),
  log10_p = -log10(df_final_pep_cta$Abundance.Ratio.Adj..P.Value...JJ012..INF....JJ012..NoINF.))
df_jj_cta <- na.omit(df_jj_cta)

# Categorise CTA peptides
df_jj_cta$category <- ifelse(df_jj_cta$log2_ratio > 1 & df_jj_cta$log10_p >
  -log10(0.05), "Upregulated", ifelse(df_jj_cta$log2_ratio <
  -1 & df_jj_cta$log10_p > -log10(0.05), "Downregulated", "Not significant"))
table_sig <- table(df_jj_cta$category)

# Volcano plot
p <- ggplot(df_jj_cta, aes(x = log2_ratio, y = log10_p)) + geom_point(aes(color = category),
  alpha = 0.7, size = 1, position = position_jitter(width = 0.1,
  height = 0.1)) + scale_color_manual(values = c(Upregulated = "red",
  Downregulated = "blue")) + geom_vline(xintercept = c(-1,
  1), linetype = "dashed") + geom_hline(yintercept = -log10(0.05),
  linetype = "dashed", color = "black") + labs(x = "Log2 Ratio",
  y = "-log10(adj. p-value)", color = "") + theme_minimal(base_size = 13) +
  theme(legend.position = "right") + annotate("text", x = 3,
  y = max(df_jj_cta$log10_p, na.rm = TRUE) + 2, label = table_sig["Upregulated"],
  hjust = 1, vjust = 1, size = 4) + annotate("text", x = -3,
  y = max(df_jj_cta$log10_p, na.rm = TRUE) + 2, label = table_sig["Downregulated"],
  hjust = 0, vjust = 1, size = 4)
```

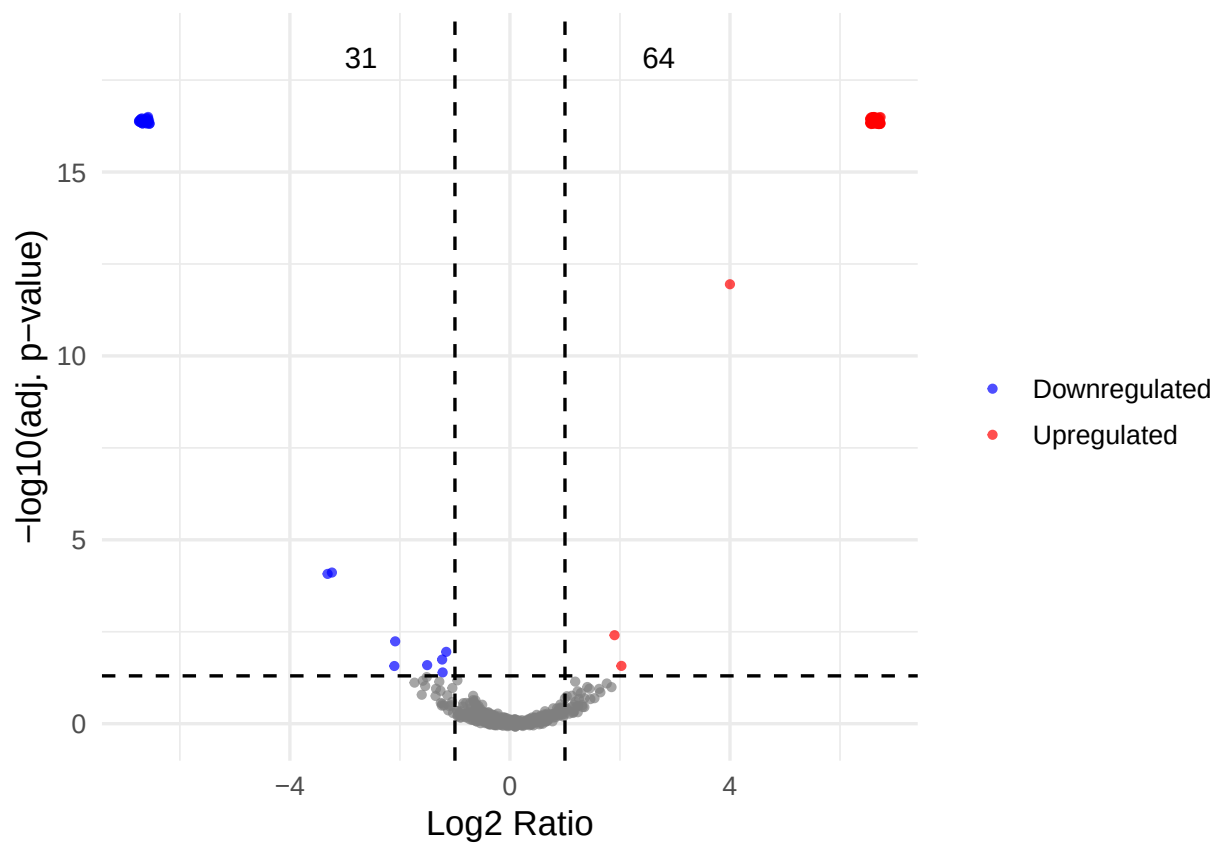


Figure 4: Volcano plot CTAs JJ012

```
arrange(desc(log2_ratio))

df_up_jj$Peptide <- gsub("^.*\\.\\.(.*)\\.\\..*$", "\\1", df_up_jj$pep)
df_up_jj$pvalue <- 10^(-df_up_jj$log10_p)
df_up_jj$Gene <- df_up_jj$Genes

df_up_jj <- df_up_jj[, c("Gene", "Peptide", "log2_ratio", "pvalue")]
```

III. Venn diagrams

```
# Extract UniProt IDs from IEDB parent molecule IRIs and
# map to gene names
df_iedb_mhc1$ID <- gsub("http://www.uniprot.org/uniprot/", "",
  df_iedb_mhc1$Molecule.Parent.IRI)
df_iedb_mhc1$ID <- sub("-.*", "", df_iedb_mhc1$ID)

up <- UniProt.ws(taxId = 9606)
gene_mapping <- AnnotationDbi::select(up, keys = unique(df_iedb_mhc1$ID[df_iedb_mhc1$ID !=
  ""]), columns = "gene_primary", keytype = "UniProtKB")
colnames(gene_mapping) <- c("From", "ID", "Gene.Names..primary.")
df_iedb_mhc1_name <- merge(df_iedb_mhc1, gene_mapping, by.x = "ID",
  all.x = TRUE)

# Keep only IEDB epitopes from CTA genes
df_iedb_mhc1_cta <- df_iedb_mhc1_name[df_iedb_mhc1_name$Gene.Names..primary. %in%
  1 CTA, ]
```

```
# Clean IEDB peptide sequences (strip modifications)
pep_iedb_mhc1_cta <- unique(df_iedb_mhc1_cta$Name)
pep_iedb_mhc1_cta <- gsub(".*|\\|+.*", "", pep_iedb_mhc1_cta)
pep_iedb_mhc1_cta <- unique(pep_iedb_mhc1_cta)

# Identify novel peptides (not in IEDB, not a substring of
# any IEDB peptide)
new_pep_ch_mhc1 <- setdiff(df_up_ch$Peptide, pep_iedb_mhc1_cta)
# Check whether any novel peptide is a substring of an IEDB
# peptide
grep_ch <- sapply(new_pep_ch_mhc1, function(x) any(grepl(x, pep_iedb_mhc1_cta)))
new_pep_ch_mhc1 <- new_pep_ch_mhc1[!new_pep_ch_mhc1 %in% names(grep_ch)[grep_ch]]
df_ch_new_pep_mhc1 <- df_up_ch %>%
  filter(Peptide %in% new_pep_ch_mhc1) %>%
  distinct(Peptide, .keep_all = TRUE)

# JJ012
new_pep_jj_mhc1 <- setdiff(df_up_jj$Peptide, pep_iedb_mhc1_cta)
grep_jj <- sapply(new_pep_jj_mhc1, function(x) any(grepl(x, pep_iedb_mhc1_cta)))
new_pep_jj_mhc1 <- new_pep_jj_mhc1[!new_pep_jj_mhc1 %in% names(grep_jj)[grep_jj]]

df_jj_new_pep_mhc1 <- df_up_jj %>%
  filter(Peptide %in% new_pep_jj_mhc1) %>%
```

```
distinct(Peptide, .keep_all = TRUE)

# Extended IEDB sets (original + substring matches)
pep_iedb_extended_ch <- unique(c(pep_iedb_mhc1_cta, names(grep_ch)[grep_ch]))
pep_iedb_extended_jj <- unique(c(pep_iedb_mhc1_cta, names(grep_jj)[grep_jj]))
```

JJ012

```
# Venn diagram: overlap between IFNg-upregulated CTA
# peptides and IEDB reference epitopes
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_jj),
  area2 = length(df_up_jj$Peptide), cross.area = length(df_up_jj$Peptide) -
  length(new_pep_jj_mhc1), category = c("IEDB", "JJ012"),
  fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)
```

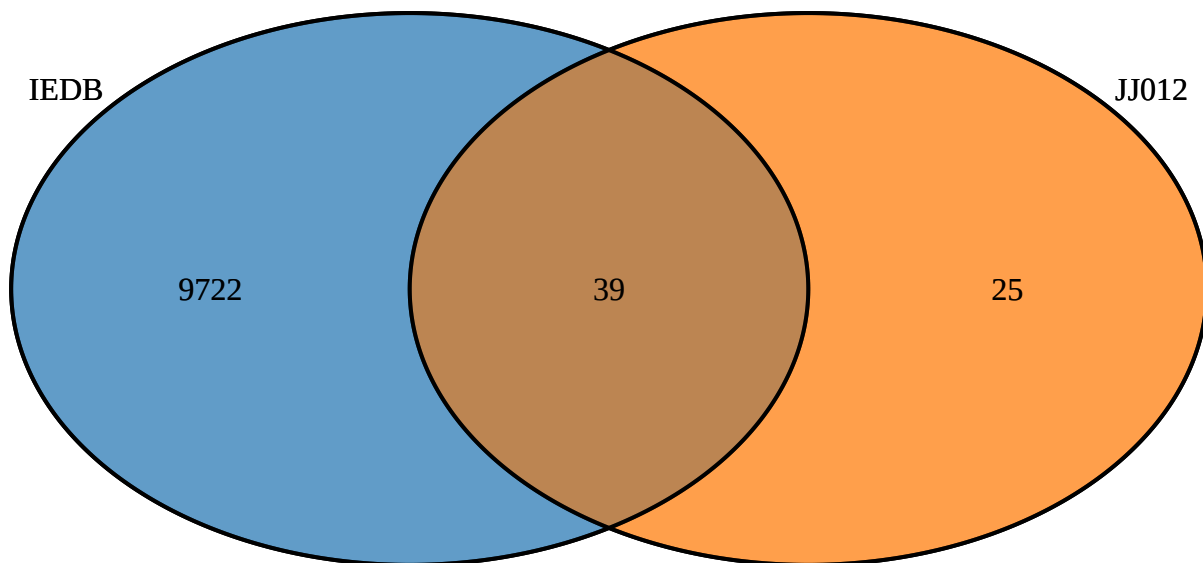


Figure 5: Venn plot JJ012 (MHCI)

CH2879

```
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_ch),
  area2 = length(df_up_ch$Peptide), cross.area = length(df_up_ch$Peptide) -
  length(new_pep_ch_mhc1), category = c("IEDB", "CH2879"),
  fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)
```

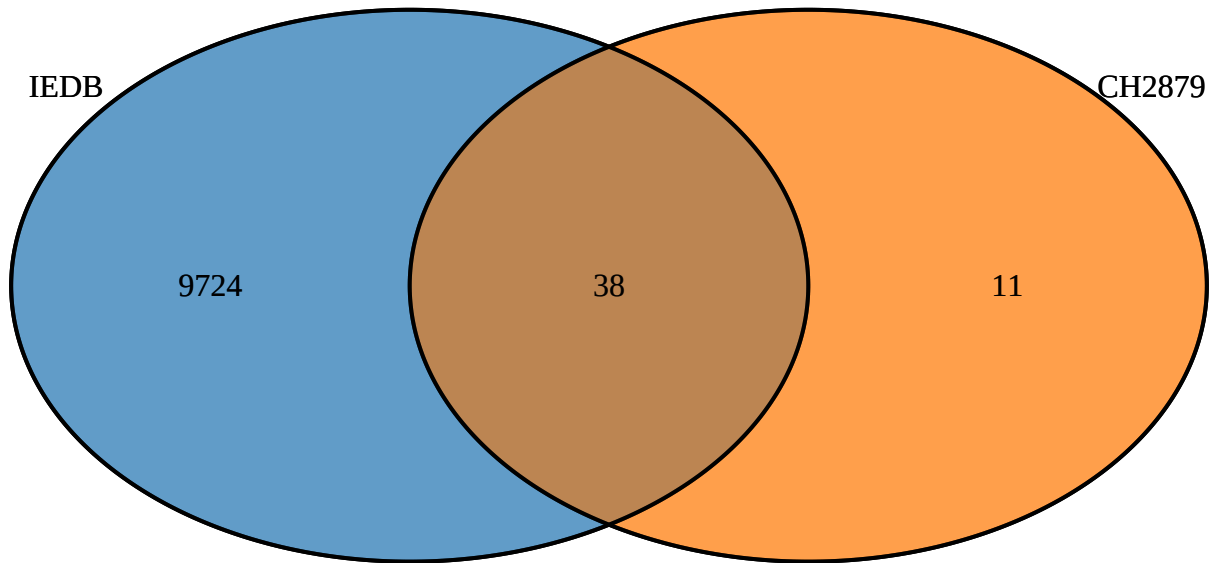


Figure 6: Venn plot CH2879 (MHCI)

IV. HLA binding affinity

```
# Keep only predictions matching upregulated CTA peptides
df_aff_ch_hlaa0101 <- df_aff[df_aff$Allele == "HLA-A0101" & df_aff$Peptide %in%
  df_up_ch$Peptide, c("Peptide", "Affinity", "Rank")]

df_aff_jj_hlaa0201 <- df_aff[df_aff$Allele == "HLA-A0201" & df_aff$Peptide %in%
  df_up_jj$Peptide, c("Peptide", "Affinity", "Rank")]
```

CH2879

```
# Identify known peptides for CH2879 (in IEDB or substring
# matches)
pep_commons <- intersect(df_up_ch$Peptide, pep_iedb_mhc1_cta)
pep_commons2 <- names(grep_ch)[grep_ch]
pep_known_all <- union(pep_commons, pep_commons2)
```



```

# Combine novel and known peptides for annotation
df_iedb_new_pep_ch <- df_up_ch %>%
  filter(Peptide %in% new_pep_ch_mhc1 | Peptide %in% pep_known_all) %>%
  distinct(Peptide, .keep_all = TRUE)

# Merge with affinity predictions
df_pep_ch_hlaa0101 <- df_up_ch[df_up_ch$Peptide %in% df_aff_ch_hlaa0101$Peptide,
]
df_pep_ch_hlaa0101 <- merge(df_pep_ch_hlaa0101, df_aff_ch_hlaa0101,
  by = "Peptide")

df_pep_ch_hlaa0101 <- df_pep_ch_hlaa0101 %>%
  mutate(Status = case_when(Peptide %in% new_pep_ch_mhc1 ~
    "New", Peptide %in% pep_known_all ~ "Known"))

# Count peptides per Rank bin and flag ranks containing
# novel peptides
df_count <- df_pep_ch_hlaa0101 %>%
  ungroup() %>%
  mutate(Status = as.character(Status)) %>%
  group_by(Rank) %>%
  summarise(nb_pep = n(), is_new = any(Status == "New"))

p <- ggplot(df_count, aes(x = nb_pep, y = Rank, color = is_new)) +
  geom_point(size = 2) + scale_color_manual(values = c(`FALSE` = "grey30",
    `TRUE` = "#ff7f0e"), labels = c(`FALSE` = "Known peptides",
    `TRUE` = "Novel peptides")) + geom_hline(yintercept = c(0.5,
    2), linetype = "dashed", color = "black") + annotate("text",
    x = max(df_count$nb_pep), y = 0.5 * 0.8, label = "Strong binders",
    hjust = 1, color = "black") + annotate("text", x = max(df_count$nb_pep),
    y = 2 * 0.8, label = "Weak binders", hjust = 1, color = "black") +
  labs(x = "CH2879 HLA-A*01:01 peptides number", y = "%Rank",
    color = "Category") + scale_y_log10(breaks = c(0.01,
    0.1, 1, 10, 100), labels = c("0.01", "0.1", "1", "10", "100")) +
  scale_x_continuous(breaks = seq(0, max(df_count$nb_pep),
    by = 1)) + theme_minimal()
p

# ggsave('../results/scatter_ch_a01.svg', plot = p)

```

JJ012

```

# Identify known peptides for JJ012
pep_commons <- intersect(df_up_jj$Peptide, pep_iedb_mhc1_cta)
pep_commons2 <- names(grep_jj)[grep_jj]
pep_known_all <- union(pep_commons, pep_commons2)

# Combine novel and known peptides for annotation
df_iedb_new_pep_jj <- df_up_jj %>%

```

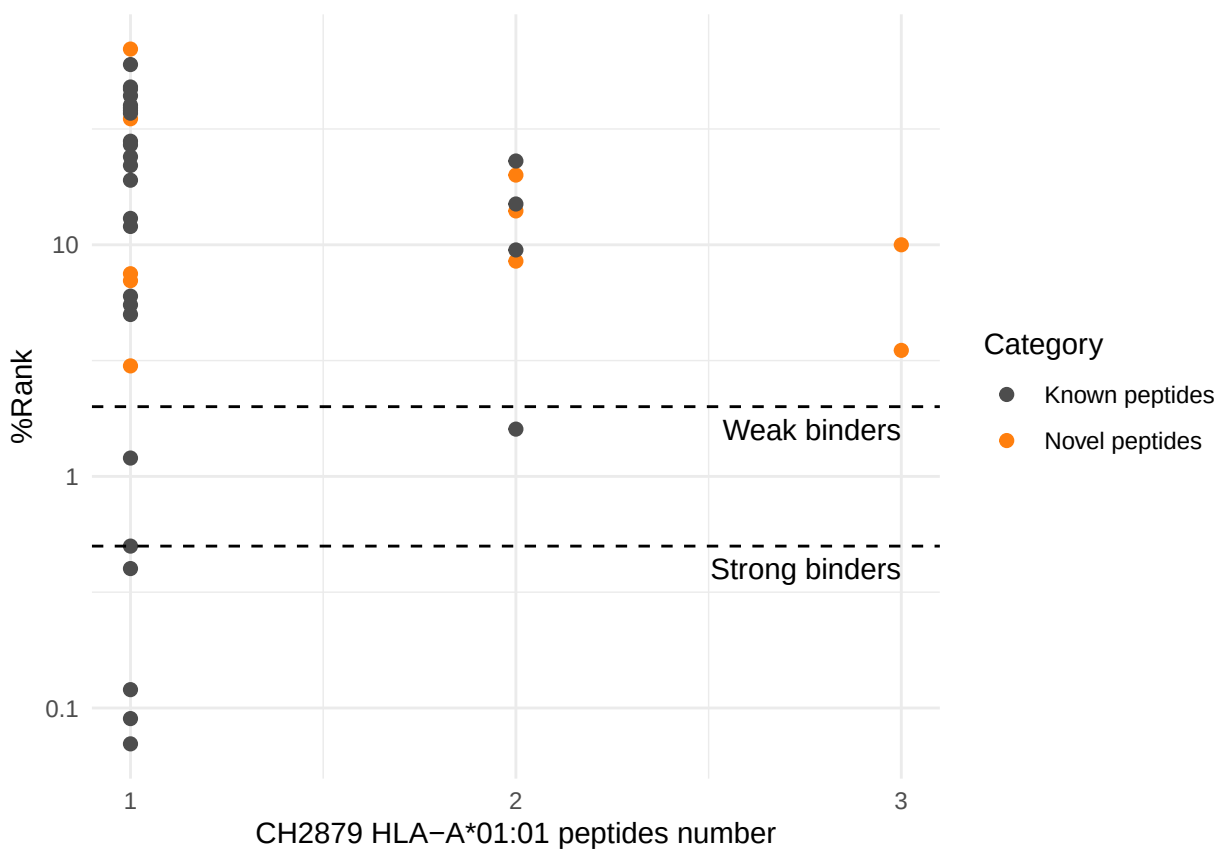


Figure 7: Scatter plot Rank CH2879

```

filter(Peptide %in% new_pep_jj_mhc1 | Peptide %in% pep_known_all) %>%
distinct(Peptide, .keep_all = TRUE)

# Merge with affinity predictions
df_pep_jj_hlaa0201 <- df_up_jj[df_up_jj$Peptide %in% df_aff_jj_hlaa0201$Peptide,
]
df_pep_jj_hlaa0201 <- merge(df_pep_jj_hlaa0201, df_aff_jj_hlaa0201,
by.x = "Peptide", by.y = "Peptide")

df_pep_jj_hlaa0201 <- df_pep_jj_hlaa0201 %>%
mutate(Status = case_when(Peptide %in% new_pep_jj_mhc1 ~
"New", Peptide %in% pep_known_all ~ "Known"))

# Count peptides per Rank
df_count <- df_pep_jj_hlaa0201 %>%
ungroup() %>%
mutate(Status = as.character(Status)) %>%
group_by(Rank) %>%
summarise(nb_pep = n(), is_new = any(Status == "New"))

p <- ggplot(df_count, aes(x = nb_pep, y = Rank, color = is_new)) +
geom_point(size = 2) + scale_color_manual(values = c(`FALSE` = "grey30",
`TRUE` = "#ff7f0e"), labels = c(`FALSE` = "Known peptides",
`TRUE` = "Novel peptides")) + geom_hline(yintercept = c(0.5,
2), linetype = "dashed", color = "black") + annotate("text",
x = max(df_count$nb_pep), y = 0.5 * 0.8, label = "Strong binders",
hjust = 1, color = "black") + annotate("text", x = max(df_count$nb_pep),
y = 2 * 0.8, label = "Weak binders", hjust = 1, color = "black") +
labs(x = "JJ012 HLA-A*02:01 peptides number", y = "%Rank",
color = "Category") + scale_y_log10(breaks = c(0.01,
0.1, 1, 10, 100), labels = c("0.01", "0.1", "1", "10", "100")) +
scale_x_continuous(breaks = seq(0, max(df_count$nb_pep),
by = 2)) + theme_minimal()

p

# ggsave('../results/scatter_jj_a0201.svg', plot = p)

```

V. Supplementary data 3

```

write_xlsx(list(JJ012_all_upregulated = df_jj_all_up_table, CH2879_all_upregulated = df_ch_all_up_table,
JJ012_CTA_upregulated = df_pep_jj_hlaa0201, CH2879_CTA_upregulated = df_pep_ch_hlaa0101),
path = "../results/fig7_supp.xlsx")

```

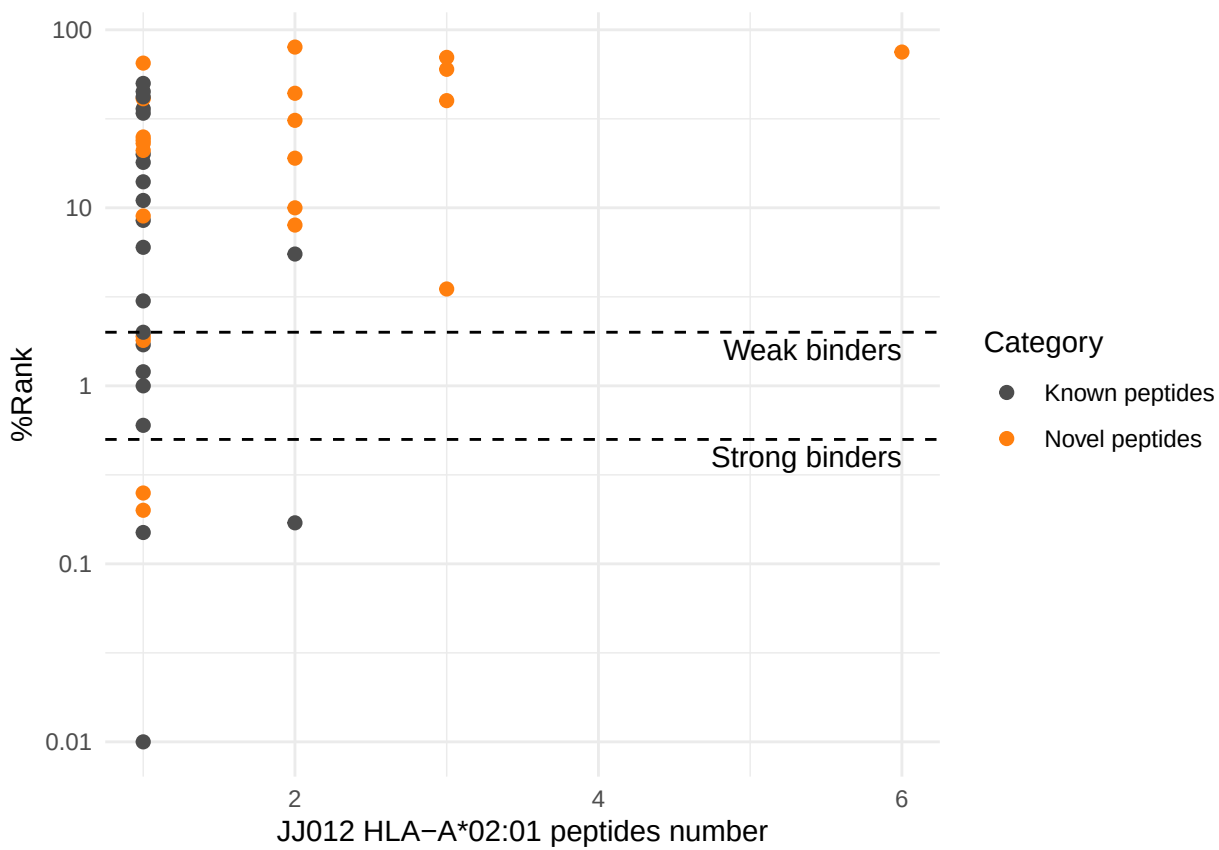


Figure 8: Scatter plot Rank JJ012